

Topic 9 — Forces and Their Effects

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

9.1 Vectors and Resultant Forces

Forces are **vectors** — they have magnitude and direction. When adding, take direction into account.

Resultant force



$$\text{Resultant} = 8 - 3 =$$

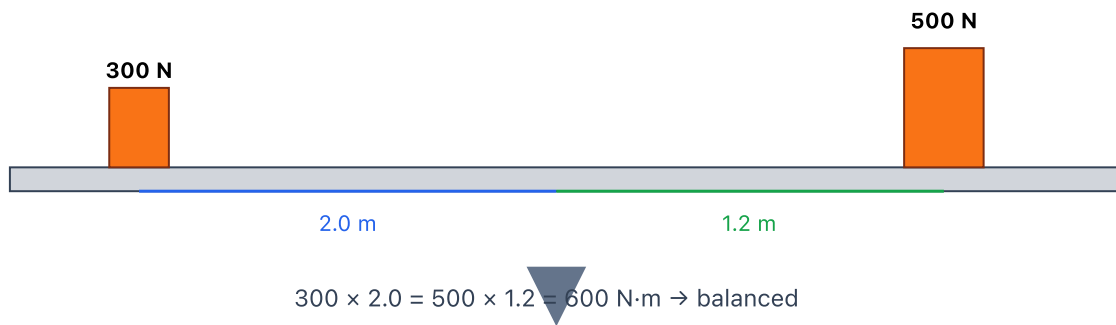
5 N right 4 N 3 N Resultant = $\sqrt{(3^2 + 4^2)} = 5 \text{ N}$

9.2 Moments, Levers and Gears

Moment of a force = **F × perpendicular distance from the pivot**. Unit: newton-metre (N·m).

Principle of moments: sum of clockwise = sum of anticlockwise (for equilibrium)

See-saw / Lever



Gears

- Small gear driving a large gear: speed decreases, **torque increases**.
- Used in bikes (low gear = high torque for hills) and in cars (gearboxes).

9.3 Centre of Mass

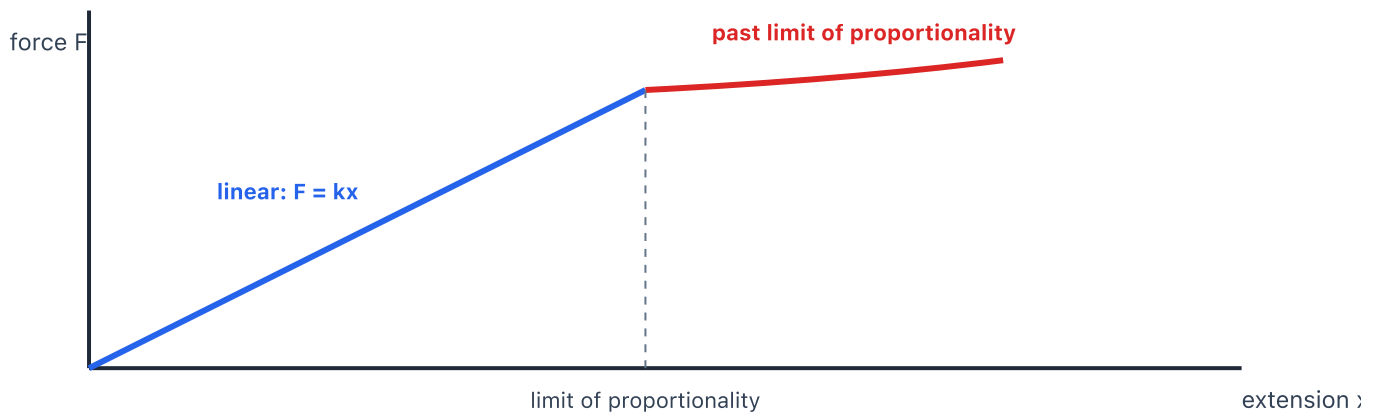
The single point where we consider an object's weight to act. Found experimentally for flat shapes using a plumb line (hang from two points; centre of mass is at the intersection of the two drawn lines).

9.4 Hooke's Law

$$F = k \times x \quad (k = \text{spring constant, N/m})$$

Valid up to the **limit of proportionality**. Beyond the **elastic limit**, the spring is permanently deformed (inelastic deformation).

Force–Extension Graph



$$\text{Elastic PE} = \frac{1}{2} \times k \times x^2$$

Key Points Summary

- Forces are vectors; combine by considering direction.
- Moment = $F \times \text{perpendicular } d$; balanced when CW = ACW moments.
- Gears trade speed for torque — used in cars and bikes.
- Hooke's law: $F = kx$ up to limit of proportionality; elastic PE = $\frac{1}{2}kx^2$.